Sylvester: A Benchmark and Method for Lifecycle-Aware Entity Resolution on Open Financial Data

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ABSTRACT

We introduce Sylvester, a public benchmark for *lifecycle-aware* entity resolution on financial entities: matching records across mergers, rebrands, spinoffs, and jurisdiction migrations where the surface form has deliberately changed. Built from public GLEIF Level-1/2, SEC EDGAR, and Wikidata, it provides 125,519 labeled pairs over 3.3M entities across three held-out evaluation regimes (time, event type, jurisdiction). Our central finding: a 22M-parameter MiniLM cross-encoder fine-tuned on this data reaches 0.83–0.87 AUROC across all three held-out regimes and 57–61 % Hits@1 on the time and event splits (±2.0–2.3 pp 95 % CI), 27–35 pp above the strongest bi-encoder and above all five zero-shot frontier LLMs tested (four organizations) on those splits. The lift concentrates on M&A, absorption, and breakup events, where the post-event name comes from the acquirer and string similarity has nothing to grip onto.

Two caveats bound this result. The benchmark deliberately mines name-similar hard negatives against name-different positives; the near-chance AUROC of most similarity baselines on the time and event splits is a construction choice, not a claim about the utility of name matching in ER generally — an *inverted*-similarity baseline exploits it for 64–70 % Hits@1 on the curated pools yet collapses off-pool and fielded; only the cross-encoder is strong across all conditions. In a fielded retrieve-then-rerank simulation over the full 3.3M-entity universe, end-to-end Hits@1 falls to 15–29 % because bi-encoder Recall@20 caps at 33–50 %; the bottleneck is the first stage, not the reranker. Code, splits, and trained weights reproduce on one MacBook in ∼2 hours.

KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Every fintech engineer, quant researcher, and compliance analyst has a story of a pipeline silently breaking on one of these transitions. The records look completely different on the surface — different legal names, different tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

Pair-matching ER — including LLM-based methods that achieve 99% F1 on point-in-time records [8] — treats each record as a snapshot of identity. Temporal entity-linking benchmarks such as TempEL [10] address mention-to-KB disambiguation against an evolving knowledge base, a different task. Financial KG construction [1] works at a single time slice. To the best of our knowledge no public benchmark targets the specific question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

A natural concern: the benchmark is engineered against name similarity by construction — negatives are mined name-similar while lifecycle successors are, by definition, name-different. This is deliberate, documented (§3.4, §5.4), and operationalized: we report the inverted-similarity exploit as its own baseline. The question the benchmark asks is the discriminative sub-problem a two-stage pipeline faces after a name-aware retriever narrows the field: *given a name-similar candidate pool, what architecture correctly identifies the true lifecycle successor?*

Contributions.

(1) **An open benchmark** (§3): 125,519 labeled pairs over 3.3M GLEIF Level-1 entities, hard negatives in four classes backed by 474K GLEIF Level-2 relationships, and three held-out regimes (time, event type, jurisdiction). Construction choices and their scope of generalizability are documented throughout (§3, §7).
(2) **A diagnostic finding** (§5): most similarity-based methods score near-chance AUROC on the time and event splits because negatives are mined to be name-similar while positives are name-different by construction—not a claim about ER generally. The construction cuts both ways: *inverted* name similarity reaches 0.64–0.70 Hits@1 on those splits, yet collapses on the TF-IDF pool, the fielded pool, and renames (§5.4). A 22M-parameter cross-encoder is the only method strong across all of these conditions (0.83–0.87 AUROC; 24–35 pp Hits@1 over the best bi-encoder), with gains concentrated on M&A and absorption events where the post-event name comes from the acquirer. The advantage survives a mechanically different miner (−2–13 pp but +13–39 pp above zero-shot on all cross-pool cells; §5.6).
(3) **The deployment bottleneck is retrieval, not reranking** (§5.6): in a fielded simulation over 3.3M entities, end-to-end Hits@1 falls to 15–29 % because bi-encoder Recall@20 caps at 33–50 %. The cross-encoder reranks effectively (46–58 % conditional Hits@1 when given the right candidates); the system breaks at the first stage, not the second.

A prompt-enrichment and larger-backbone (12-layer, L-12) ensemble extension further lifts Hits@1 by 10–11 pp on time/event (Table 2); the headline result compared against baselines uses the single fine-tuned 6-layer (L-6) cross-encoder.

2 RELATED WORK

Entity resolution and record linkage. Classical ER [5] matches records via similarity over structured fields. Modern PLM cross-encoders on serialized records (GraLMatch [4]) are now standard, and the OpenSanctions Pairs corpus [8] benchmarks LLM matching at 755K pairs. All adopt a *snapshot* framing, not lifecycle-spanning matches.

Temporal entity linking. TempEL [10] time-stratifies entity linking against an evolving Wikipedia, but the task is mention-to-KB disambiguation, not record-to-record matching over legal entities with structured lifecycle events.

Financial entity knowledge graphs. Recent work at the ACM Conference on AI in Finance constructs financial-filing knowledge graphs (FinReflectKG [1]) at a single time slice; the lifecycle-bridge question is orthogonal.

3 THE SYLVESTER BENCHMARK

3.1 Data sources

We construct Sylvester entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF concatenated file.** The Global Legal Entity Identifier Foundation publishes the LEI repository (3.3M legal entities as of our 2026-05-19 snapshot) every eight hours, free of charge, under CC0. Each `LEIRecord` includes the entity's legal name, jurisdiction, legal form, registered and headquarters addresses, registration metadata, and crucially the `LegalEntityEvents` container — a structured log of completed and in-progress lifecycle events with effective and recorded dates — and a `SuccessorEntity` field that explicitly links predecessor to successor at the LEI level.
- **GLEIF Level-2 Relationship Records.** 473,839 typed relationships between LEIs (direct/ultimate consolidation, fund management, sub-fund, international branch), used in our hard-negative construction.
- **Wikidata.** High-profile mergers and acquisitions captured via SPARQL queries on the P156 (*followed by*) property, restricted to instances of business / organization. Used as a marquee qualitative supplement and for cross-validation.
- **SEC EDGAR.** The 10,354-record `company_tickers.json` file gives a free name ↔ CIK crosswalk that maps to GLEIF for the U.S. public-company subset (3,334 confident name-match links).

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element; of these, 33,645 have a populated `SuccessorLEI` (the remaining 16,557 reference a successor entity that never registered an LEI — useful for entity-death modeling but excluded from training).

Label quality validation. A five-class auto-classifier on the successor pairs (using documented event types, normalized name delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428 (75.6%); `likely_real_name_changed` 3,330 (9.9%); `likely_real_event_only` 1,704 (5.1%); `ambiguous` 1,355 (4.0%); **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%), yielding 31,817 plausibly-real predecessors (31,824 positive pairs; some predecessors have multiple successors).

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 RR edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction: coincidental name collision vs. true lifecycle linkage.
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication: 253,733 candidate (predecessor, hard-negative) pairs covering 20,138 of 31,817 predecessors (~63%). The remaining 37% are small entities in jurisdictions without name collisions; they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type-priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test); Figure 1 shows the underlying event-volume distribution. Test composition: 3,448 positives, >2,000 anchors with at least one positive.
- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption. The test set is heavily dominated by *absorption* (2,609 of 2,760 anchors, 94.5%); demerger (120), breakup (25), and spinoff (6) are present but statistically thin; per-event-type metrics within the split are reported in §5.
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 88 anchors, 161 positives — the smallest split; its headline numbers carry ~±8.5 pp 95% bootstrap CIs (vs. ~±2.0–2.3 pp on time/event).

Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

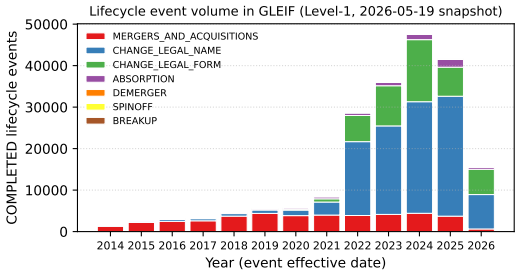


Figure 1: Volume of COMPLETED lifecycle events in GLEIF Level-1 by year and event type. The 2022 cliff reflects GLEIF tightening event-reporting requirements; pre-2022 data is sparser and reporting-biased.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 both-sides-resolved to GLEIF, 283 not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Wyeth → Pfizer, Interbrew → AB-InBev); not used in the headline evaluation.

4 METHOD

We present two methods at different cost-accuracy points: a bi-encoder (§4.1) that embeds each side independently and scores by cosine — cheap, but hemmed in by the anti-correlation property — and a cross-encoder (§4.2) that reads both records together, 4–10× more expensive per pair, producing the headline result.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ `|` `jurisdiction:` c_ℓ `|` `form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard neg as negatives), 3 epochs, batch 32, AdamW lr= 2×10^{-5}, 100 warmup, ~6 min on MPS. Ablation variants in Table 1: `method_no_hard` (in-batch negs only), `method_v2_full` (adds E5-cosine hard negs + 1,372 transitive positives). Loss is contrastive; the cross-encoder below uses BCE on the logit — different objectives by architectural convention, not strictly controlled.

4.2 Cross-encoder

A bi-encoder must place each record at a fixed point in embedding space, yet the event generating each positive deliberately changes the record's surface form — it is asked to keep cosine-close two records that look very different on the page. A cross-encoder reads both records jointly and can compare specific tokens between them.

We fine-tune a second-stage cross-encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. The backbone is the 22M-parameter `ms-marco-MiniLM-L-6-v2` cross-encoder from HuggingFace; training uses the same positive set as the bi-encoder (including transitives) plus a 4:1 sample of E5-cosine hard negatives per positive, with positives labeled 1.0 and negatives 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps. The extensions in §5.3 hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

Inference cost. Scoring all 125,519 pairs in the benchmark takes ≈80 seconds on MPS. In deployment the cross-encoder would re-rank a retriever's top-K (the standard two-stage IR pattern); the benchmark's fixed 5–10-negative pools make direct full scoring feasible, and Table 1 reflects that setting.

Hard-negative re-mining. The initial token-overlap fallback (92% of negs) injected enough label noise that explicit hard-negs slightly underperformed in-batch alone. Replacing it with E5-cosine top-20 same-country candidates yields 652,764 total negs across 31,714 predecessors (vs. 253,733 before; coverage ~20K → 31,714) and erases the "hard-negs hurt" anomaly for the bi-encoder, while noticeably helping the cross-encoder.

5 EXPERIMENTS

5.1 Baselines

We compare against four similarity baselines: `exact_name` (normalized legal-name equality), `token_set` (Jaccard over name tokens), `tfidf_char` (char-3gram TF-IDF cosine), and `sentence_encoder` (zero-shot E5-small-v2 cosine — the method's backbone without fine-tuning). We also report six zero-shot LLM pairwise baselines under one protocol: one API call per pair, byte-identical prompts, a 0–10 integer rubric, `temperature=0` where exposed. Five models spanning four organizations run at *full* test coverage (26,564 pairs each): **Claude Haiku 4.5**, **Sonnet 4.6**, **Opus 4.7**, **Gemini 3.1 Pro**, **DeepSeek V4 Pro** (thinking disabled); **gpt-oss-20b** runs locally on a 56/52/50-anchor sample (Table 1 footnotes).

Paired permutation tests on per-anchor expected Hits@1 separate the comparisons cleanly. The single L-6 cross-encoder beats all five API models on TIME and EVENT at $p < 10^{-4}$ (+15 to +37 pp). On the 88-anchor JURIS split it leads Haiku, Gemini, and DeepSeek ($p \le 0.018$), is within noise of Sonnet ($p = 0.14$), and is statistically indistinguishable from Opus 4.7, which holds the higher point estimate (0.767 vs. 0.750, $p = 0.74$). The L-6×L-12 ensemble beats Opus on TIME/EVENT ($p < 10^{-4}$); JURIS is a coin flip (0.784 vs. 0.767). Zero-shot quality is strikingly vendor-heterogeneous: every Claude model outscores both other vendors' frontier models on every split — including Haiku, the cheapest model tested (§7).

5.2 Metrics

We report per-anchor retrieval metrics (Hits@1, MRR) computed by ranking each anchor's candidates by score, and AUROC for the pair-wise classification framing; Hits@3, AUPRC, and threshold-tuned F1 ship with the released per-run metrics. Discrete scorers tie frequently within a pool, so Hits@k/MRR are exact *expectations over uniformly random tie-breaking* (a fixed tie order is biased).

5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the E5-neg evaluation pool

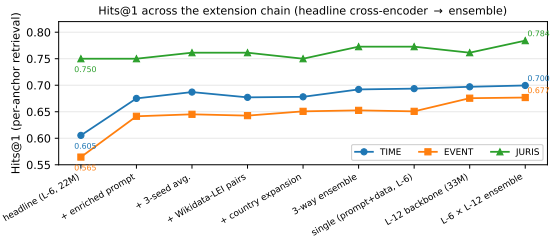


Figure 2: Hits@1 across the extension chain. Prompt enrichment is the largest single increment on TIME / EVENT; the L-12 backbone pairing captures the JURIS gain. JURIS CIs (±8.5 pp, $n = 88$) overlap across the chain; read its per-step deltas as directional.

(`pairs.parquet`). Headline cross-encoder Hits@1 95% bootstrap confidence intervals are ±2.0 pp (TIME), ±2.3 pp (EVENT), and ±8.5 pp (JURIS, $n = 88$ anchors); the jurisdiction-split gain comfortably clears its CI band but absolute numbers on that split should be treated as directional. We retest the headline cross-encoder on a mechanically different candidate pool (§5.6) and find the gap shrinks but does not collapse; Table 2 and Figure 2 extend the L-6 cross-encoder to the L-6×L-12 ensemble.

Extensions: enriched prompt + L-6×L-12 ensemble. Stacking four modifications on the headline cross-encoder — prompt enrichment (GLEIF Level-2 parent/child names, founding year, ISO country expansion, category), Wikidata-LEI mining (259 GLEIF positives recovered, 49 cross-jurisdiction), multi-seed averaging, and ensembling with an L-12-backbone variant — lifts Hits@1 a further +10 to +11 pp on time/event (+3 on jurisdiction, inside its CI band; Table 2, Figure 2). The ensemble reaches 0.700 / 0.677 / 0.784 Hits@1; the headline result reported against baselines remains the single L-6 cross-encoder, so that comparison stays apples-to-apples. A negative result is worth recording: symmetric scoring (averaging $s(a, b)$ and $s(b, a)$) hurts −0.9/−1.9/−9.1 pp, so the cross-encoder really does learn a directional predecessor→successor representation.

Backbone size does not break the bi-encoder ceiling. Scaling E5-small (33M) to E5-base (110M) leaves TIME/EVENT Hits@1 unchanged (−0.2/−1.2 pp) and lifts JURIS +2.3 pp: the constraint is structural, not parameter-bound — cosine-close embeddings for lexically-different pairs are not a problem more parameters fix.

5.4 The construction cuts both ways: an anti-similarity baseline

`token_set`, `tfidf_char`, and `sentence_encoder` all land below 0.5 AUROC on the held-out-time and held-out-event-type splits (Table 1). The cause is mechanical: hard negatives are deliberately name-similar to the predecessor while positives are by construction name-different, so on this candidate distribution name similarity is *evidence against* a true successor link. So we report the mirror image as its own baseline: `anti-token_set` ranks candidates by *negated* Jaccard, and on the curated pool it is strong — 0.700 / 0.638 / 0.516 Hits@1, *above* the single L-6 cross-encoder on time and event — because picking the least name-similar candidate exploits the construction.

The exploit does not survive any change of conditions. On Sylvester-TF-IDF it collapses to chance (0.25 / 0.25 / 0.21 Hits@1, AUROC ≈ 0.5) while the cross-encoders hold 0.53–0.66 (Table 4); on the fielded pool it drops to 9–18 % conditional Hits@1 vs. 46–58 % for the cross-encoder; per event it is a one-way bet on acquirer-renaming — 0.72 on M&A but 0.10 on pure renames and 0.35 on legal-form changes, where the cross-encoder scores 0.90 and 0.85. The cross-encoder also wins pairwise AUROC on every split (0.83–0.87 vs. 0.68–0.81) and Hits@1 on JURIS (+25 pp), and the L-6×L-12 ensemble beats the exploit on all three splits. Curated-pool Hits@1 alone can therefore be gamed; the cross-encoder's claim is being strong *simultaneously* on the curated pool, the alternative-miner pool, the fielded pool, and all event types. The mechanism is joint attention: asking whether the *differences* between two records look like a lifecycle-event pattern, not how similar two strings are.

5.5 Where the lift is real: per-event-type breakdown

Aggregate Hits@1 hides large per-event-type variation. The cross-encoder does real work on M&A, absorption, and breakup events, where the post-event name comes from the acquirer rather than the target and string-similarity cannot bridge the gap. On pure renames it does essentially nothing: those are already at 0.90 Hits@1 zero-shot.

On the held-out-time split (Figure 3), the cross-encoder's gain is concentrated in the operationally hard events. Pure renames (`CHANGE_LEGAL_NAME`, $n = 10$) are essentially solved by every method including zero-shot E5-small (0.90 Hits@1, no lift). The headline gains come from `MERGERS_AND_ACQUISITIONS` ($n = 1{,}506$: 0.18 zero-shot → 0.62 L-6 → 0.71 ensemble) and `ABSORPTION` ($n = 461$: 0.23 → 0.53 → 0.65), where the post-event name comes from the acquirer. The ensemble also more than doubles zero-shot on the small but operationally critical `BREAKUP` subset ($n = 7$: 0.14 → 0.71).

On the held-out-event split, "held-out event type" really means "held-out absorption." The held-out-event test set is 94.5% `ABSORPTION` anchors (§3.4); within the Hits@1-eligible subset, `DEMERGER`, `BREAKUP`, and `SPINOFF` contribute $n = 42$, $n = 20$, and $n = 4$ respectively. We report Table 3 so that the contribution of each held-out type is visible separately. The cross-encoder lift is consistent across the absorption-like types (`ABSORPTION`: 0.22 → 0.57, `DEMERGER`: 0.29 → 0.55, `BREAKUP`: 0.15 → 0.40); the `SPINOFF` cell is statistically meaningless at $n = 4$. The aggregate "+34 pp Hits@1 over zero-shot E5 on the event split" number is therefore primarily a transfer claim about absorption, not a strong claim about spinoffs.

5.6 Robustness to candidate-pool shift

The headline cross-encoder result is trained and (mostly) evaluated against candidates from the same family of hard-negative miners — v1 token-overlap and v2 E5-cosine both rank candidates by name-related signals, and the cross-encoder trains on E5-cosine negs. The obvious question: does the gap survive an evaluation pool built

Table 1: Test-set metrics on the three split regimes. Bold = headline L-6 cross-encoder used in baseline comparisons; the larger L-12 matches or beats it on every metric, and the L-6×L-12 ensemble (Table 2) is the overall best. Random ranking scores 0.18–0.22 Hits@1 (pools average 7.4–8.3 candidates). The best fine-tuned bi-encoder lifts Hits@1 6–8 pp over zero-shot E5-small; the cross-encoder adds a further 24–35 pp and is the only method above 0.8 AUROC on both time and event (every direct-similarity, LLM, and bi-encoder row stays at or below 0.55 there). `anti-token_set` ranks by *negated* similarity — strong on this constructed pool, collapsing off-pool (§5.4). Hits@1/MRR are expectations over uniformly random tie-breaking. LLM rows: one call per pair, byte-identical prompts, full test coverage (26,564 pairs), `temperature=0` where exposed. On JURIS, Opus 4.7's 0.767 exceeds the L-6's 0.750 (n.s., p=0.74). See Table 4 for candidate-pool shift. Hits@1 95% CIs: ±2.0 / ±2.3 / ±8.5 pp. †Opus: no temperature param; Gemini (preview): thinking not disableable, lowest level, Batch API. ‡local; 56/52/50-anchor sample.

	Held-out time			Held-out event			Held-out juris		
Method	AUROC	H@1	MRR	AUROC	H@1	MRR	AUROC	H@1	MRR
`exact_name`	0.474	0.163	0.373	0.523	0.193	0.400	0.571	0.352	0.539
`token_set`	0.193	0.136	0.319	0.266	0.180	0.356	0.323	0.356	0.522
`tfidf_char`	0.277	0.180	0.360	0.349	0.213	0.386	0.568	0.420	0.583
`sentence_encoder`	0.315	0.204	0.385	0.386	0.232	0.408	0.648	0.432	0.587
`anti-token_set` (1−Jaccard)	0.807	0.700	0.761	0.734	0.638	0.715	0.677	0.516	0.618
LLM pairwise, zero-shot — uniform single-pair protocol, full test coverage									
`Claude Haiku 4.5`	—	0.290	0.476	—	0.335	0.517	—	0.618	0.738
`Claude Sonnet 4.6`	—	0.301	0.492	—	0.364	0.542	—	0.672	0.758
`Claude Opus 4.7`†	—	0.350	0.534	—	0.416	0.589	—	0.767	0.839
`Gemini 3.1 Pro`†	—	0.235	0.433	—	0.278	0.474	—	0.569	0.699
`DeepSeek V4 Pro`	—	0.260	0.451	—	0.309	0.497	—	0.585	0.712
`gpt-oss-20b`‡	—	0.185	0.377	—	0.203	0.403	—	0.523	0.683
`method_full` (bi-enc, token-overlap negs)	0.403	0.255	0.425	0.451	0.291	0.450	0.667	0.500	0.627
`method_no_hard` (bi-enc, no explicit negs)	0.407	0.259	0.429	0.461	0.299	0.459	0.668	0.511	0.641
`method_v2_full` (bi-enc, E5-cosine negs)	0.405	0.250	0.424	0.457	0.296	0.455	0.694	0.500	0.629
Cross-encoder (L-6, 22M)	**0.836**	**0.607**	**0.725**	**0.825**	**0.569**	**0.701**	**0.868**	**0.750**	**0.818**
Cross-encoder (L-12, 33M)	0.844	0.631	0.742	0.836	0.598	0.724	0.870	0.750	0.823

Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]. Headline L-6 row repeated from Table 1; intermediates plotted in Figure 2.

Model	TIME	EVENT	JURIS
Cross-encoder (L-6, 22M)	.605 [.586,.627]	.565 [.542,.589]	.750 [.659,.830]
+ enriched prompt	.675 [.653,.697]	.641 [.620,.662]	.750 [.659,.830]
+ Wikidata-LEI positives	.694 [.669,.712]	.651 [.630,.673]	.773 [.682,.853]
+ L-12 backbone (33M)	.697 [.678,.716]	.676 [.657,.697]	.761 [.670,.841]
Ensemble L-6×L-12	**.700 [.681,.718]**	**.677 [.656,.700]**	**.784 [.682,.864]**

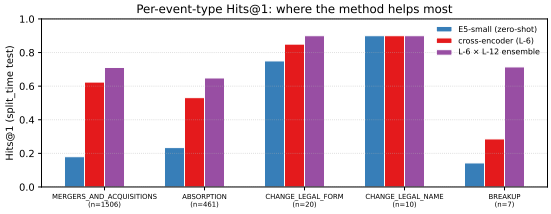


Figure 3: Per-event-type Hits@1 on the held-out-time test split: zero-shot E5-small vs. the L-6 cross-encoder vs. the L-6×L-12 ensemble. Renames (n = 10) are nearly solved by every method; the ensemble's gains over the L-6 concentrate on M&A (+9 pp) and absorption (+12 pp).

with a mechanically different miner? If yes, the cross-encoder advantage is a property of the method; if no, the headline number is partly an artifact of how the negatives were chosen.

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine (same signal as the `tfidf_char` baseline) — chosen because it has no learned representation and shares no parameters with E5. The other three classes (parent/subsidiary, same-name same/diff-country) are held constant. We call the resulting pair set SYLVESTER-TF-IDF; splits and positives are unchanged. We train two cross-encoders: one on the original E5-cosine hard negatives (trn-E5), one on the TF-IDF negatives (trn-TF-IDF), and score each on both pair pools. The 2×2 train/evaluate matrix has diagonal cells in-distribution, cross-pool cells out-of-distribution at the candidate-pool level.

What the 2×2 tells us. First, the cross-encoder *architecture* generalizes well across miners: the TF-IDF-trained cross-encoder evaluated on the E5-neg pool reaches 0.588 / 0.533 / 0.750 Hits@1 — within 1–2 pp of the E5-trained model on TIME and EVENT (surpassing it by 4.5 pp on the thin JURIS split, consistent with its wide CI), despite never having seen E5-mined candidates during training. The result is symmetric: the E5-trained model on the TF-IDF pool likewise stays well above any non-cross-encoder method.

Second, there is a real but bounded in-distribution effect. The E5-trained cross-encoder loses 2–13 pp Hits@1 when moved to the

Table 3: Per-anchor-event-type Hits@1 on the held-out-EVENT test split (within the Hits@1-eligible subset, i.e., anchors whose gold successor appears in the candidate pool), for the zero-shot baseline (E5-small) and the L-6 cross-encoder. The full test split is 94.5% `ABSORPTION` (2,609 of 2,760); `SPINOFF` at $n = 4$ is statistically uninformative. Lift sources are concentrated in the absorption-like types.

Event type	n anchors	zero-shot E5	Cross-enc (L-6)	Δ
ABSORPTION	1,546	0.223	**0.568**	+34.5 pp
DEMERGER	42	0.286	**0.548**	+26.2 pp
BREAKUP	20	0.150	**0.400**	+25.0 pp
SPINOFF	4	0.500	0.250	−25.0 pp[†]

[†] $n = 4$, statistically uninformative.

TF-IDF evaluation pool (0.600 → 0.545 on time; 0.705 → 0.571 on jurisdiction). Conversely, the TF-IDF-trained model gains +7–11 pp on the TF-IDF pool over the E5-trained one. So training on the same miner used to construct the eval candidates is worth roughly 2–13 pp Hits@1 — non-trivial, but not the entire headline gap.

Third, the cross-encoder gain over zero-shot E5 *is robust* to the miner if we are willing to retrain the cross-encoder: the diagonal cells (in-distribution) show +20 to +40 pp gains across all six splits. The cross-pool cells (out-of-distribution at training time) show +13 to +39 pp gains — smaller, but still well outside any plausible CI band. Joint encoding does real work on lifecycle ER — but the absolute size of the gain depends on how the candidate pool was mined, so the "+27–35 pp" figure in the abstract should be read in that light.

Fielded retrieve-then-rerank. The miner-robustness experiment above shifts the mining signal but still draws candidates from a curated negative pool. A real deployment pipeline does something different: a first-stage retriever pulls the top-K candidates from the full entity universe, then a second-stage model reranks. We simulate that here. The trained `method_v2_full` bi-encoder retrieves top-20 candidates per test anchor from the full 3.3M-entity GLEIF universe, restricted to the anchor's jurisdiction (the lightest filter a production system would apply); the cross-encoder reranks. The filter itself is nearly free: only 1.8–4.1 % of test positives have a successor in a different country, so it caps recall at ≥96 % — far above the observed 33–50 %; the bottleneck is the embedding, not the filter.

Bi-encoder Recall@20 is 33.2 / 39.6 / 49.7% on TIME / EVENT / JURIS. That is the hard ceiling for any reranker. When the bi-encoder does retrieve the true successor, the L-6 cross-encoder picks it #1 with 46–58% Hits@1 and the L-6×L-12 ensemble reaches 53–60% — both rerankers re-score the actual fielded candidates, and the ensemble gain over the L-6 cross-encoder is modest once the bi-encoder distractors look genuinely lifecycle-plausible. End-to-end deployment Hits@1 (recall × conditional, un-retrieved truths counted as misses) is 15–29% for the L-6 cross-encoder and 17–30% for the ensemble. The drop from 57–61% in-distribution on time/event (75% on JURIS, $n = 88$) to 15–30% deployment confirms rather than contradicts §4.1: the bi-encoder's fixed-embedding limit, which fails to retrieve lifecycle-spanning successors by cosine similarity, is exactly the bottleneck the anti-correlation analysis predicts.

5.7 Ablations

State features help modestly. Ablating jurisdiction and legal form from the prompt costs 1.5–2 pp Hits@1 on every split; the country token contributes most on held-out-jurisdiction, least on held-out-event-type.

Explicit hard negatives slightly *hurt* in v1 (Table 1). `method_no_hard` edges out the v1 full model by 0.4–1.1 pp Hits@1 on every split. The culprit is the `token_overlap_same_country` class, 92% of all mined v1 hard negatives and the lowest-precision class — the fuzzy fallback injects label noise, and in-batch negatives from MultipleNegativesRanking are cleaner on their own. This motivates the v2 switch from token-overlap to E5-cosine mining, and pushes back on the default assumption that more hard negatives is always better.

5.8 Additional diagnostics

Four diagnostics, compressed to fit the page budget; full tables and figures are in the released code. **B-cubed cluster F1** [2] on the test partition is 0.92 / 0.90 / 0.91 for the cross-encoder vs. ≤0.47 for any bi-encoder (each of which collapses to one giant cluster at the tuned threshold). Restricting to records in true clusters of size ≥ 2 — 64% of test records are singletons — gives a more honest 0.85 / 0.84 / 0.88. **Calibration and selective prediction.** ECE (10-bin) is 0.20, Brier 0.18: the model is over-confident in its high-score bin. A train-fitted temperature ($T \approx 1.5$) trims Brier only marginally and does not reduce ECE, so we report uncalibrated scores. A selective-prediction [6] sweep on the ensemble hits Hits@1 ≥ 0.94 at 20% coverage and 0.87–0.97 at 50% across the three splits; users who need high precision can just threshold on the score (curve in repo). **Cross-domain transfer.** Zero-shot Magellan Walmart-Amazon: the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC. The fine-tuned representation is specialized to lifecycle-spanning legal-entity matching, not generic ER. **Label quality, second opinion.** A rule-based re-classifier — orthogonal to the primary's documented-event-type evidence, though still name-informed (name-stem similarity, cross-script checks, legal-form continuity) — agrees with the primary auto-classifier on 98.7% [95% CI 97.0–100%] of the 231/277 stratified-sampled positives it judges confidently; the remaining 46 (16.6%) are marked uncertain. This bounds heuristically detectable label noise below 2% there; human expert review is future work.

5.9 Where each method is fooled

The negative-class breakdown (Table 5) reports per-negative-class mean score: higher means the method is more often fooled by that class of negative. The fine-tuned bi-encoders roughly halve the zero-shot E5 baseline's score on `parent_subsidiary` and `same_name_diff_country` negatives, but `same_name_same_country` remains a wall at ≈0.90 mean score: the bi-encoder cannot reliably tell a real successor link from two distinct entities sharing an exact normalized name in the same country.

Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 train/evaluation matrix. "E5-neg pool" = `pairs.parquet` (original benchmark pool, dominant negative class is E5-cosine top-20); "TF-IDF pool" = `pairs_v3.parquet` (TF-IDF char-3gram cosine, no neural representation). Row labels indicate which pool's hard negatives were used for cross-encoder training. E5-small zero-shot is included as a reference baseline on each pool. The cross-encoder advantage is large and consistent when train and evaluation pools match (diagonal: +20 to +40 pp); the advantage shrinks but stays substantial cross-pool (+13 to +39 pp). Numbers reproduce: original-paper Cross-encoder (L-6) predictions and re-trained `trn-E5` cells agree within 2 pp on time/event splits.

	Eval on E5-neg pool (`pairs.parquet`)			Eval on TF-IDF pool (`pairs_v3.parquet`)		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.203	0.234	0.432	0.265	0.319	0.441
	±1.8	±2.1	±10.5	±1.4	±1.7	±7.8
cross-enc trn-E5	**0.600**	**0.551**	0.705	0.545	0.528	0.571
	±2.2	±2.4	±8.8	±1.8	±1.7	±7.5
cross-enc trn-TF-IDF	0.588	0.533	**0.750**	**0.655**	**0.614**	**0.640**
	±2.0	±2.4	±9.1	±1.5	±1.7	±7.1
Δ *vs. E5, in-distribution*	+39.7	+31.7	+27.3	+39.0	+29.5	+19.9
Δ *vs. E5, out-of-distribution*	+38.5	+29.9	+31.8	+28.0	+20.9	+13.0

Table 5: Negative-class mean score (lower is better). *p/s* = parent/subsidiary, *sn-D* = same-name/diff-country, *sn-S* = same-name/same-country, *tok* = token-overlap same-country. The ensemble improves three classes but *worsens p/s* (0.70 → 0.78) — prompt-injected L2 context makes parent/subsidiary pairs look more similar, not less.

Method	p/s	sn-D	sn-S	tok
`exact_name`	0.013	1.000	1.000	0.000
`token_set`	0.144	1.000	1.000	0.486
`tfidf_char`	0.227	0.630	0.804	0.590
`sentence_encoder`	0.834	0.921	0.963	0.905
`method_full`	0.454	0.454	0.899	0.736
`method_v2_full`	0.457	0.464	0.899	0.736
Cross-encoder (L-6, 22M)	0.696	0.237	0.452	0.278
Ensemble (L-6×L-12)	0.781	**0.160**	**0.345**	**0.254**

The cross-encoder breaks that wall: `same_name_same_country` mean score drops from 0.90 (best bi-encoder) to 0.45, and the L-6×L-12 ensemble drops it further to 0.34. The ensemble also nearly solves `same_name_diff_country` (0.92 zero-shot → 0.16) and `token_overlap_same_country` (0.91 → 0.25). The one regression is `parent_subsidiary` (0.45 bi-encoder → 0.70 L-6 cross-encoder → 0.78 ensemble): the joint encoder occasionally treats parent–subsidiary edges as positive lifecycle bridges, and the ensemble's prompt enrichment that injects GLEIF L2 parent/child names makes the regression *worse*, not better. With only 221 examples in this class the headline impact is small, but the trend points at a clear direction: explicit multi-class modeling of corporate-hierarchy relationships.

6 ANALYSIS

6.1 Error analysis

We sampled 100 `method_full` failures on the held-out-time test split (positives not ranked first) and categorized them via simple text features: 66 mixed/other, 19 where anchor and true successor share *no* tokens, 8 with substring overlap with a wrong candidate, and 7 exact normalized-name collisions with a wrong candidate.

The 19 "no shared tokens" cases are the hardest: the true successor of a name like `Nima N.V.` (NL) — here `Beheermaatschappij Arros B.V.` — shares zero meaningful tokens with either the predecessor or the wrong top-1 candidate `NIMA B.V.`. No similarity-based representation can solve these; they require world knowledge or explicit graph context (board overlap, regulatory filings, press releases). One representative failure: anchor `DB HR Solutions GmbH` (DE), true successor `DEUTSCHE BANK AKTIENGESELLSCHAFT`, top-1 wrong `HR Solutions GmbH` — resolving "DB" to Deutsche Bank is external knowledge a 33M-parameter encoder is unlikely to memorize from name strings alone.

Residual JURIS failures cluster on sibling consolidation. Of the 19 / 88 anchors the ensemble mis-ranks on JURIS, ≥11 are sibling consolidations (e.g., 9 "`EPIC <port> PTE. LTD.`" Singapore ships into `BW EPIC KOSAN VESSELS ONE`; the model returns sibling `EPIC` vessels); a second cluster is Japanese-script name changes the ASCII tokenizer cannot bridge (Daiwa-Sumigin → Mitsui-Sumitomo DS). An LLM with multilingual pretraining and ownership-graph context would address both.

6.2 Why this matters in practice

The benchmark targets a failure mode common across financial infrastructure. Stripe trains transaction and merchant embeddings at scale [9]; Ramp's accounting agents flag duplicate vendors and invoices [7]; Bloomberg's entity backbone underpins its NLP and financial modeling work [3]. Each of these pipelines faces the lifecycle-ER gap: entity records that look completely different after a merger, rebrand, or absorption. Sylvester gives practitioners a public benchmark to evaluate whether their system handles it.

7 LIMITATIONS AND FUTURE WORK

Six limitations; longer versions ship with the released code.

The "anti-correlation" finding is a construction artifact. Hard negatives in Sylvester are mined to be name-similar to the predecessor; positives are by definition name-different. The near-chance

AUROC of most similarity baselines (§5) follows directly from that mining design and does not generalize to ER outside this benchmark. The finding motivates joint encoding here; it is not a universal claim about name similarity in ER.

In-distribution headline overstates deployment. The headline cross-encoder Hits@1 (57–61 % on time/event; 75 % on JURIS, $n = 88$, directional) is on the E5-neg mined candidate pool; on a fielded retrieve-then-rerank pipeline that draws candidates from the full 3.3M-entity GLEIF universe (§5.6), end-to-end Hits@1 falls to 15–29 % (L-6 cross-encoder) / 17–30 % (ensemble). The cause is upstream: bi-encoder Recall@20 is 33–50 %, so the reranker often never sees the true successor. The miner-robustness experiment shows the cross-encoder gain itself is partly in-distribution but not purely a within-miner artifact (+13 to +39 pp cross-pool). A stronger first-stage retriever — a learned hybrid sparse-dense index, or a coarsely pre-filtered cross-encoder shortlist — is the highest-leverage target for future work.

Seed variance and a sliver of mislabeled hard negatives. Bi-encoder Hits@1 over seeds 42/7/1337 was 0.251 ± 0.005 (TIME), 0.291 ± 0.004 (EVENT), 0.523 ± 0.016 (JURIS); TIME/EVENT are stable, JURIS varies ±1.6 pp consistent with its small test sample ($n = 88$). Cross-encoder multi-seed retraining (Table 2) shows similar behavior. Separately, 205/93,695 = 0.22% of hard negatives appear in GLEIF's `SuccessorEntity`; the cleaned variant `pairs_clean.parquet` drops them with ≤0.2 pp impact.

Reporting / survivorship bias + thin slices. GLEIF `SuccessorEntity` is voluntarily populated, biasing positives toward well-counseled regulated entities. The held-out-event test set is 94.5% absorption (the label oversells event-type generality), and the held-out-jurisdiction test has only 88 anchors (±8.5 pp Hits@1 CI). The cross-encoder gain clears the CI band on all three splits, but per-split absolute numbers should be read with those sample sizes in mind. A 277-pair second-classifier check (§5.8) bounds detectable label noise at <2% on the confident-positive subset; expert human review would give a stronger bound. Splits are pair-level, not entity-disjoint (32 % / 21 % of time-split test candidates / gold successors appear in training pairs; JURIS <1 %), though L-6 Hits@1 shows no seen-entity advantage (unseen 0.62 vs. seen 0.57 TIME; 0.56 vs. 0.58 EVENT, both n.s.); entity-disjoint splits are a stricter future design.

LLM baselines: what the discrete scale hides. Discrete elicitation caps what ranking can express: models forfeit 13–48 pp Hits@1 to within-pool score ties. Gemini 3.1 Pro scores every candidate in a pool *identically* on 55% of TIME anchors — oracle tie-breaking would give it 0.72 vs. 0.23 measured — while Sonnet/Opus collapse only ∼5% of pools; the cross-encoder's continuous scores pay no tie tax. Finer elicitation (wider scales, log-probs, comparative tie-breaks) may recover part of this. Prompt format also moves results materially (batched multi-candidate scoring: Sonnet 0.43 TIME Hits@1 vs. 0.30 single-pair), and cross-vendor gaps may partly reflect differential training exposure to GLEIF-derived text. The cost gap stands: 0.64 ms/pair after one-time training vs. ∼1 s/pair metered.

The model is specialized, and parent/subsidiary is a thin slice. The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Walmart-Amazon, −5.9 pp AUROC; §5.8); the fine-tuned representation is specialized to lifecycle ER, not generic. The parent/subsidiary negative class shows a regression (mean score 0.45 → 0.70 from bi-encoder to L-6 cross-encoder, worsening to 0.78 in the ensemble), but with only 221 negatives the effect-size estimate is not statistically meaningful. Explicit multi-class modeling of related-but-distinct entities is a clear direction for future work.

8 CONCLUSION

Sylvester is an open benchmark for lifecycle-aware entity resolution on financial entities, built from public GLEIF, SEC EDGAR, and Wikidata data. Hard negatives are mined to be name-similar to each predecessor and true successors are by construction name-different, so similarity-based baselines stay at or below 0.55 AUROC on the time and event splits — a property of this benchmark's construction, not of name similarity in ER generally (inverted similarity scores 0.64–0.70 Hits@1 on-pool, collapsing off-pool; §5.4). A 22M-parameter MiniLM cross-encoder reaches 0.83–0.87 AUROC across all three splits and 57–61 % Hits@1 on the time and event splits (75 % on held-out jurisdiction, $n = 88$, directional), 24–35 pp above the strongest bi-encoder and above all five zero-shot frontier LLMs tested on time/event ($p < 10^{-4}$); on jurisdiction only Opus 4.7 matches it (0.767 vs. 0.750, n.s.). Re-mining negatives shrinks the advantage 2–13 pp but keeps it +13 to +39 pp above zero-shot; a fielded retrieve-then-rerank simulation over 3.3M entities drops end-to-end Hits@1 to 15–30% because bi-encoder Recall@20 caps at 33–50%. The real gains localize to M&A, absorption, and breakup; pure renames are already at 0.90 Hits@1 zero-shot.

Reproducibility. Code, parsers, predictions, and trained weights: https://github.com/chang-07/er-research; end-to-end repro from raw GLEIF ∼2 hours on one MacBook.

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